

ENGINEERING PHYSICS
SUBJECT CODE- NBS-101/NBS-201

COURSE OBJECTIVES: Understanding of the principle and concepts of Quantum Mechanics, Electromagnetic field theory, wave optics, Lasers, fiber optics, nano materials and superconductors for their applications in Engineering.

SYLLABUS:

Content	Contact Hours
Unit-1: Quantum Mechanics	9
Inadequacy of classical mechanics, Planck's theory of black body radiation(qualitative), Compton effect, de-Broglie concept of matter waves, Davisson and Germer Experiment, Phase velocity and group velocity, Time-dependent and time-independent Schrodinger wave equations, Physical interpretation of wave function, Particle in a one-dimensional box and Quantization of energy.	
Unit-2: Electromagnetic Field Theory	8
Basic concept of Stoke's theorem and Divergence theorem, Basic laws of electricity and magnetism, Continuity equation for current density, Displacement current, Maxwell equations in integral and differential form, Maxwell equations in vacuum and in conducting medium, Poynting vector and Poynting theorem, Plane electromagnetic waves in vacuum and their transverse nature. Relation between electric and magnetic fields of an electromagnetic wave, Plane electromagnetic waves in conducting medium, continuity conditions of electric field and magnetic field, Skin depth.	
Unit-3: Wave Optics	10
Coherent sources, Division of wave front, division of amplitude, Interference in uniform and wedge shaped thin films, Necessity of extended sources, Newton's Rings and its applications, Introduction to diffraction, Fraunhofer diffraction at single slit and double slit, Absent spectra, Diffraction grating, Dispersive power, and Resolving power of grating, Rayleigh's criterion of resolution, Spectra with grating.	
Unit-4: Fiber Optics & Laser	9
Fibre Optics: Principle and construction of optical fiber, Acceptance angle, Numerical aperture, Acceptance cone, Step index and graded index fibers, Fiber optic communication principle, Attenuation, Dispersion, Application of fiber. Laser: Absorption of radiation, Spontaneous and stimulated emission of radiation, Population inversion, Method of achieving populations inversion, Einstein's Coefficients, Principles of laser action, Solid state Laser(Ruby laser) and Gas Laser (He-Ne laser), Laser applications.	
Unit-5: Superconductors and Nano-Materials:	8

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